Aquaculture Analysis

EDS 223: Assignment 3

Monique Hernandez

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Background

Marine aquaculture have the potential to be a more sustainable option than the alternative land-based meat for the global food supply (Hall et al, 2011). Gentry et al. created a map based on the neccessay constraints for marine aquaculture including, ship traffics, dissolved oxygen, and bottom depth, concluding that the global seafood demand could be met with less than 0.015% of the global ocean area.

Assignment Description

For this assignment, the Exclusive Economic Zone (EEZ) will be used to determine suitable regions throughout the west coast to develop marine aquaculture for several oyster species, as well as a species of choice, which was chosen to be the blue mussel (*Mytilus edulis*). Blue mussels re eurythermal and well adapted to temperatures of 5–20 °C, with an upper thermal tolerance limit of about 29 °C for adults (Matoo et al., 2021).

The main goal is to create a function that will input the following: minimum and maximum sea surface temp, minimum and maximum depth, and the species name. The output will be a map of the EEZ regions colored by the amount of suitable areas for the selected species.

For the oysters, the information is the following:

- **Oysters:**
  - Sea surface temperature: 11-30°C
  - Depth: 0-70 meters below sea level
- **Blue mussel:**
  - Name: *Mytilus edulis*

- Sea surface temperature: 5-20°C
- Depth: 0-60 m

Data Preparations

```
# Load libraries
library(tidyverse)
library(sf)
library(here)
library(stars)
library(tmap)
library(janitor)
library(terra)
library(geodata)
library(kableExtra)
```

```
# Read in data for the average SST
set_file <- c(
    here('data/average_annual_sst_2008.tif'),
    here('data/average_annual_sst_2009.tif'),
    here('data/average_annual_sst_2010.tif'),
    here('data/average_annual_sst_2011.tif'),
    here('data/average_annual_sst_2012.tif')
)

# File path for SST data stack
sst_data <- rast(set_file)

# Load bathymetry data
depth <- rast('data/depth.tif')

# Load West Coast EEZ
eez_data <- st_read('data/wc_regions_clean.shp')
```

Data Process

```
# Calculate SST mean from 2008-2012 and convert Kelvin to Celsius
sst_mean_c <- mean(sst_data, na.rm = TRUE) - 273.15
```

```r
# Crop data to match
crop_depth <- crop(depth, sst_mean_c)

# Transfer sst data using bilear method to match
resampled_depth <- resample(crop_depth, sst_mean_c, method = "bilinear")
```

```r
# Verify resolution
  # Checking and converting depth to match SST resolution, extent, and CRS
  if (all(res(sst_mean_c) == res(resampled_depth))) {
    print("Resolutions match!")
  } else {
    warning("Resolutions don't match!")
    res(resampled_depth) <- res(sst_mean_c)
  }

  if (ext(sst_mean_c) == ext(resampled_depth)) {
    print("Extents match!")
  } else {
    warning("Extent don't match!")
    ext(resampled_depth) <- ext(sst_mean_c)
  }
# test if the resampled depth CRS matches the sst_mean_c CRS
  if (crs(resampled_depth) == crs(sst_mean_c)) {
    print("CRS match!")
  } else {
    warning("CRS don't match!")
    crs(resampled_depth) <- crs(sst_mean_c)
  }
```

Workflow for Oysters

```r
# variables for min and max
min_sst <- 11
max_sst <- 30

# Reclassifying SST and depth data into suitable locations
# SST reclassification matrix
  rcl_sst <- matrix(
    c(
      -Inf, min_sst, 0,
```

```
      min_sst, max_sst, 1,
      max_sst, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
  )

min_depth <- 0
max_depth <- -70

# Depth reclassification matrix
rcl_depth <- matrix(
    c(
      -Inf, max_depth, 0, # should max_depth be negative?
      max_depth, min_depth, 1,
      min_depth, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
  )
```

Find Suitable Zones for Oysters

```
# reclassify sst and depth to suitable zones
zone_sst <- classify(sst_mean_c, rcl = rcl_sst)

zone_depth <- classify(resampled_depth, rcl = rcl_depth)

# Combine to adhere to satisfy both sst and depth for suitable zones
depth_sst_zones <- c(zone_sst, zone_depth)
```

```
# Find suitable zones for oysters adhering to both conditions
suitable_zones <- lapp(depth_sst_zones,
 function(sst_data, depth_data) sst_data * depth_data)
```

Find suitable total area inside EEZ

```r
# Converting sst data to match suitable zones
sf::st_crs(eez_data)<- terra::crs(suitable_zones)
```

```r
# Convert to raster
eez_raster <- rasterize(eez_data, suitable_zones, field = "rgn_id")
```

```r
# Calculate cell size
cell_size <- cellSize(suitable_zones, unit = "km", mask = TRUE)
```

```r
# Use only the cells with value 1 as suitable in the total area
suitable_zmask <- mask(cell_size,suitable_zones, maskvalues = 0)
```

```r
# Select raster cells by suitable zpmes inside EEZ
desired_zone <- zonal(suitable_zmask,eez_raster, fun = 'sum', na.rm = TRUE)
```

Create Map for Oyster Colored by EEZ Regions

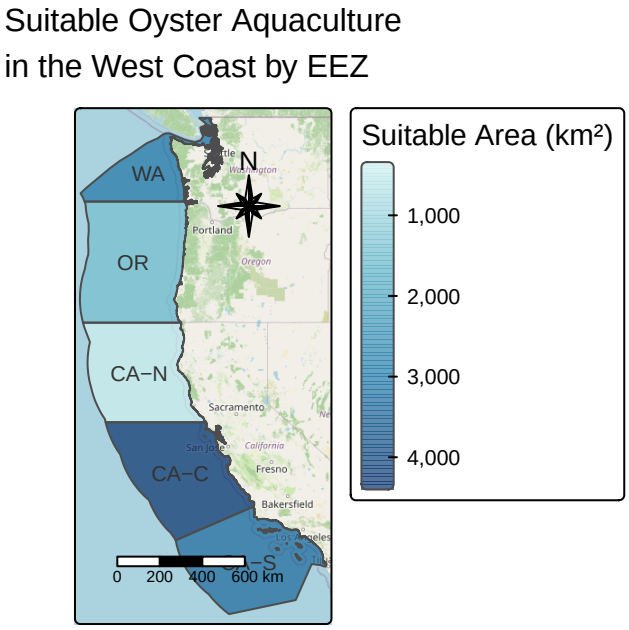
```r
# Create tibble
z_tibble <- as_tibble(desired_zone)

# Merge for desired zones to EEZ data
map_1 <- left_join(eez_data, z_tibble, by = "rgn_id")
```

```r
# Creating a color palette to differentiate the regions
color_palette <- c("#d0f0f0", "#5fb3c7", "#1f78a8", "#08306b")
```

```r
# Visualize oyster suitable area by EEZ region
tm_basemap("OpenStreetMap") +
  tm_shape(map_1) +
  tm_polygons(
    fill = "area",
    fill_alpha = 0.7,
    col = "grey30",
    fill.scale = tm_scale_continuous(values = color_palette),
    fill.legend = tm_legend(
      title = "Suitable Area (km²)",
      position = tm_pos_out("right", "center")
    )
```

```
  ) +
  tm_text(text = "rgn_key", size = 0.7, col = "grey20") +
  tm_compass(type = "8star",
        size = 2, position = c("right", "top")) +
  tm_scalebar(position = c("left", "bottom"), size = 5) +
  tm_options(component.autoscale = FALSE) +
  tm_title("Suitable Oyster Aquaculture\nin the West Coast by EEZ", size = 1)
```



Numerical Oyster Reprsentation within EEZ Regions

```
# Create a table for area by region and suitability
map_1 |>
  st_drop_geometry() |>
  select(rgn_id, area, area_km2) |>
  arrange(desc(area)) |>
  mutate(pct_suitable = area / area_km2 * 100) |>
  kable(
    digits = 1,
    col.names = c(
      "EEZ Region",
```

```
      " Suitable Area (km²)",
      "Total EEZ Area (km²)",
      "Suitable (%)"
    ),
    caption = "Oyster Aquaculture by Exclusive Economic Zone (EEZ) in the West Coast",
    booktabs = TRUE
  ) |>
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 1: Oyster Aquaculture by Exclusive Economic Zone (EEZ) in the West Coast

EEZ Region	Suitable Area (km²)	Total EEZ Area (km²)	Suitable (%)
3	4373.6	202738.3	2.2
4	3329.9	206860.8	1.6
5	3091.5	66898.3	4.6
1	1395.7	179994.1	0.8
2	323.9	164378.8	0.2

Function: Find Suitable Area for Marine Species

```
# Function
eez_species <- function(
    min_sst,
    max_sst,
    min_depth,
    max_depth,
    species_name
){
# SST matrix reclassification
    rcl_sst <- matrix(
    c(
      -Inf, min_sst, 0,
      min_sst, max_sst, 1,
      max_sst, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
    )
```

```r
# Reclassify SST
    zone_sst <- classify(sst_mean_c, rcl = rcl_sst)

# Reclassify depth
    rcl_depth <- matrix(
    c(
      -Inf, max_depth, 0,
      max_depth, min_depth, 1,
      min_depth, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
    )

    zone_depth <- classify(resampled_depth, rcl = rcl_depth)

# Combine the two zones together
    depth_sst_zones <- c(zone_sst, zone_depth)

# Find suitable zones
    suitable_zones <- lapp(depth_sst_zones,
    function(sst_data, depth_data) sst_data * depth_data)

# Determine suitable zones inside EEZ
    sf::st_crs(eez_data)<- terra::crs(suitable_zones)

# Convert to raster
    eez_raster <- rasterize(eez_data, suitable_zones, field = "rgn_id")

# Calculate cell size
    cell_size <- cellSize(suitable_zones, unit = "km", mask = TRUE)

# Keep cell area only where suitable areas equals 1

    suitable_zmask <- mask(cell_size,suitable_zones, maskvalues = 0)

# Use zonal function to sum masked result and eez raster

    desired_zone <- zonal(suitable_zmask,eez_raster, fun = 'sum', na.rm = TRUE)

# Create tibble
    z_tibble <- as_tibble(desired_zone)
```

```r
# Merge for desired EEZ regions
    map <- left_join(eez_data, z_tibble, by = "rgn_id")

# Create color palette
   color_palette <- c("#d0f0f0", "#5fb3c7", "#1f78a8", "#08306b")

# Visualize EEZ
   map_species <- tm_basemap("OpenStreetMap") +
  tm_shape(map) +
  tm_polygons(
    fill = "area",
    fill_alpha = 0.85,
    col = "grey30",
    fill.scale = tm_scale_continuous(values = color_palette),
    fill.legend = tm_legend(
      title = "Suitable Area (km²)",
      position = tm_pos_out("right", "center")
    )
  ) +
  tm_text(text = "rgn_key", size = 0.7, col = "grey20") +
  tm_compass(type = "8star", size = 2, position = c("right", "top")) +
  tm_scalebar(position = c("left", "bottom")) +
  tm_options(component.autoscale = FALSE) +
  tm_title(paste0("Suitable ", species_name,
  " Areas by EEZ in the West Coast"), size = 1)

# Create a table

table <- map |>
  select(rgn_id, area, area_km2) |>
  arrange(desc(area)) |>
  mutate(pct_suitable = area / area_km2 * 100) |>
  st_drop_geometry()

return(list(map_species, table))

}
```

EEZ Species Function: Blue Mussel

Utilizing the function created above, a map, as well as a table with the total suitable area depicted numerically, can be easily recreated for the desired blue mussel rather than recreate the lengthy process of manually inputing the inforamtion for sea surface temperature and depth.

The function is called after EEZ species and used as shown:

- `eez_species(min_sst, max_sst, min_depth, max_depth, species_name)`

Simply Input into the funtion:

- Oysters:
  - min_sst = 11°C
  - max_sst = 30°C
  - min_depth = 0 meters
  - max_depth = -70 meters
  - Species_name = Oysters
- Blue mussels:
  - min_sst = 5°C
  - max_sst = 20°C
  - min_depth = 0 meters
  - max_depth = -60 meter
  - species_name = Blue Mussel
- Output:
  - map of EEZ regions colored by amount of suitable area

```
# Verify function by using oysters
oyster <- eez_species(11, 30, 0, -70, 'Oysters')
oyster[[1]]
```

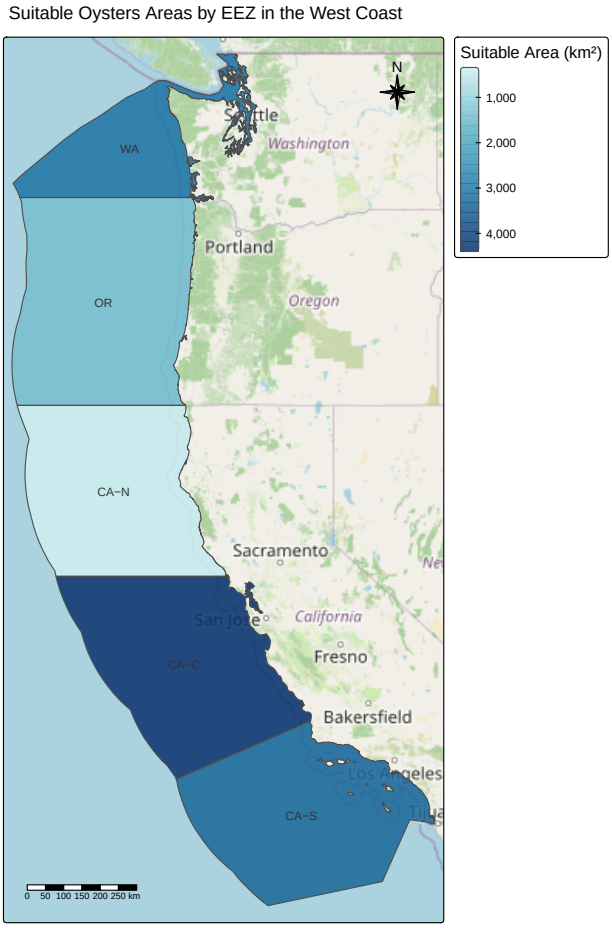


Figure 1:

Map of the West Coast showcasing the suitable areas that meet several oyster's sea surface temperature and depth for aquaculture possibilites based on Exclusive Economic Zones.

```
# Make a table for blue mussel
knitr::kable(oyster[[2]],
digits = 2, col.names = c("EEZ Region",
      " Suitable Area (km²)",
      "Total EEZ Area (km²)",
      "Suitable (%)"))
```

Table 2: Table depicting the total area (Km^2) by EEZ regions in the West Coast for several oyster species.

EEZ Region	Suitable Area (km²)	Total EEZ Area (km²)	Suitable (%)
3	4373.56	202738.33	2.16
4	3329.90	206860.78	1.61
5	3091.52	66898.31	4.62
1	1395.73	179994.06	0.78
2	323.91	164378.81	0.20

```
# Use function for blue mussel
blue_mussels <- eez_species(5, 20, 0, -60, 'Blue Mussel')
blue_mussels[[1]]
```

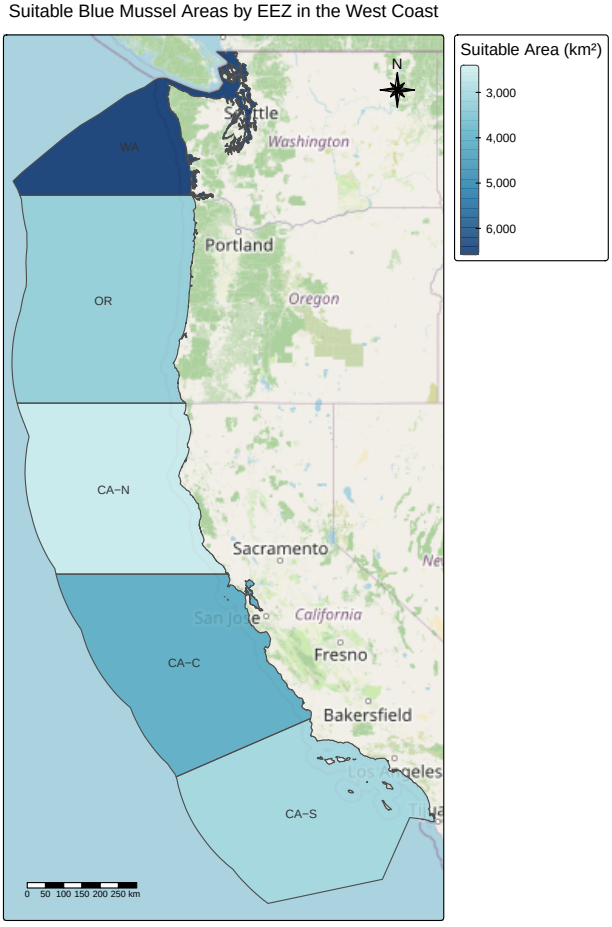


Figure 2:

Map of the West Coast showcasing the suitable areas that meet the Blue Mussel's sea surface temperature and depth for aquaculture possibilites based on Exclusive Economic Zones.

```
# Make a table for blue mussel
knitr::kable(blue_mussels[[2]],
digits = 2, col.names = c("EEZ Region",
      " Suitable Area (km²)",
      "Total EEZ Area (km²)",
      "Suitable (%)"))
```

Table 3: Table depicting the total area (Km^2) by EEZ regions in the West Coast for Blue Mussel species.

EEZ Region	Suitable Area (km²)	Total EEZ Area (km²)	Suitable (%)
5	6545.83	66898.31	9.78
3	3910.14	202738.33	1.93
1	3152.90	179994.06	1.75
4	2938.33	206860.78	1.42
2	2410.14	164378.81	1.47

Reflection

Using the Exclusive Economic Zones to map out suitable areas for marine species that have a high potential for aquaculture development in the west coast of the United States. The conditions for the species were determined by using the sea surface temperature and depth below, oyster with a range of 11-30°C and maximum depth of 70 meters below sea level, and blue mussels with a range of 5-20°C and a maximum depth of 60 meters.

Once the maps were created, once by manually coding and then again utilizing the function, the area for oyster farming was found to be in Central California with a total of 4,373.6 km^2 and the lower bound being in Northern California with 323.9 km^2.

As for the second species, the Blue Mussel's information was inputted into the function and both a map of the suitable locations were generated as well as a table. Both depict the same information with the suitable location being in Washington with a total of 6545.8 km^2 and the lowest being in Northern California with a total of 2410.1 km^2.

Data Citations

"Blue Frontiers: Managing the Environmental Costs of Aquaculture", S. J. Hall, A. Delaporte, M. J. Phillips, M. Beveridge, and M. O'Keefe, available here

"Daily Global 5km Satellite Sea Surface Temperature Anomaly (Version 3.1)", NOAA Coral Reef Watch, 2018, available here [Accessed: Nov. 7, 2026]

"GEBCO_2022 Grid", GEBCO Compilation Group, British Oceanographic Data Centre, 2022, available here [Accessed: Nov. 7, 2026]

"Maritime Boundaries and Exclusive Economic Zones", Flanders Marine Institute, MarineRegions.org, 2026, available here [Accessed: Nov. 7, 2026]

"Mytilus edulis, Blue Mussel", M. L. D. Palomares and D. Pauly (Eds.), SeaLifeBase, available here [Accessed: Nov. 7, 2026]

"Temperature but not ocean acidification affects energy metabolism and enzyme activities in the blue mussel, Mytilus edulis", O. B. Matoo, G. Lannig, C. Bock, and I. M. Sokolova, Ecology and Evolution, 2021, available here [Accessed: Nov. 7, 2026]